CREATE Module: The Mathematics of Fluid Mechanics lecture 1

1D Elasticity and deformation, Stress-strain relation

- Continuum mechanics is the study of the macroscopic motions of matter.

- To distinguish it from classical mechanics we often say that continuum mechanics is concerned with **deformation.**

- Since we will be discussing both the motion of solids (briefly) and liquids it would be more accurate to say **deformation and the rate of deformation.**

- Continuum Mechanics builds on math you may or may not have taken, but here I proceed in a fairly gentle way.

- Apart from a brief introduction, fluids will occupy the entirety of this module

- Continuum mechanics makes one key hypothesis, the so-called **Continuum Hypothesis**

- This comes in various guises, but all predicated on the fact that in any macroscopic sample of matter we have an unthinkably large number of molecules (think Avogadro's number or about 10^{23})

- May be the prettiest statement of the Continuum Hypothesis is Euler's concept of a fluid particle, or a parcel of fluid small enough to be treated as a singleton but that retains macroscopic properties like temperature, pressure, etc.

- Let's begin with some basic conservation rules, that are sometimes presented in course on PDEs

- Consider a 1D domain $[a, b]$ with a continuous concentration of a chemical of interest, $c(x, t)$. The statement for the total amount of chemical is: $C(t) = \int_a^b c(x, t) dx$

- The rate of change of the total amount of chemical is $C'(t) = \int_a^b c_t \, dx$ where I use the shorthand notation $\frac{\partial f}{\partial t} = f_t$.

- The reason I was able to pass the derivative inside the integral is because the interval in question is fixed.

- This won't be the case when we treat a moving fluid, but it's a good place to start.

- What can the rate of change with time be balanced by?

- Well first of all we could have sources (S^+) and sinks (S^-), and these need to be added up via integration:

$$\int_a^b \left[S^+(x,t) - S^-(x,t)\right] \; dx$$

- Second we could have flux into the domain at the left and right ends of the domain.

- We need to choose a direction we associate with "positive flux" and it makes to have flux to the right, or increasing x, be positive. The flux is thus: $-(Fl(b) - Fl(a))$

- I wrote it the way I did to suggest a relation to the Fundamental Theorem of Calculus

$$\int_a^b f(x) \; dx = F(b) - F(a) \text{ where } F'(x) = f(x).$$

- The total balance equation thus reads $\int_a^b c_t dx = \int_a^b S^+(x,t) - S^-(x,t)\ dx - \int_a^b Fl_x\ dx$ where I used the Fundamental Theorem of Calculus.

- We can regroup this under a single integral like this $\int_a^b [c_t - S^+(x,t) + S^-(x,t) + Fl_x]\ dx = 0$

- We haven't said anything about what a and b are so the result holds for any interval.

- A theorem called the Dubois-Reymond Lemma (proved in lecture 2) then guarantees that the argument of the integral must be zero and we get a partial differential equation: $c_t - S^+(x,t) + S^-(x,t)dx + Fl_x = 0$

- The problem with $c_t - S^+(x,t) + S^-(x,t) + Fl_x = 0$ is that it is not closed.

- We need some way to relate the flux to the concentration c. This is a science, and not a math problem.

- A very well known example is Fick's Law, which says $Fl = -\kappa c_x$ or that the flux is proportional to how quickly concentration changes in space. κ is a positive, physical constant that must be found by measurement.

- The PDE can then be rearranged so the rate of change with time on the left is balanced by the sum of three terms $c_t = \kappa c_{xx} + S^+ - S^-$

- The sources increase the concentration, the sinks shrink it, and the diffusion moves it around.

- $c_t = \kappa c_{xx} + S^+ - S^-$ does not have an exact solution, but it is nice since we understand all the terms it contains.

- This is going to be a consistent game plan; we will seek to derive equations and leave their solution to further work.

- In modern scientific work, a lot of solving is actually done numerically.

- An important question we will have to face is how to extend these derivations to higher dimensions.

- This will begin in lecture 3.

- The module is structured to provide tangible outcomes in three categories.
- First I want you to understand basic concepts.
- Second I want you to understand the difference between fluids and solids (as well as something about materials in between).
- Third, and finally, I want you to see the mathematics behind the curtain, without spending multiple courses on it.
- By the end of the module you ideally have more confidence to question the equations people present in their papers.

Elasticity in 1D

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- The silly putty pictured above is deforming due to the effects of gravity.
- We wish to come up with a description of this deformation using the abstract notion of a continuum (basically saying the putty is a subset of $\mathbf{R}^3$).
- We need a quantitative way to measure deformation and forces that are unique to a continuum.

Deformation in 1D

- Consider a metal rod which is clamped on one end and squished along its length at the other end.

- For concreteness imagine the rod initially occupies [0,L]

- After being squished it occupies a different interval, say [0,l]
- Indeed every point on the rod can be thought of as the domain of a function that maps the before (or undeformed) state to the after (or deformed) state.
- We can imagine coming with global and local (pointwise) measures of deformation, but in general we would like them all to be independent of the original length of the rod.
- Notice that we can normalize by both the initial and final states.

$$\frac{L-l}{L} \qquad \frac{L^2-l^2}{L^2} \qquad \frac{L^2-l^2}{l^2}$$

Deformation in 1D

- Local measures can be defined in three steps (see notes for derivation).
- First choose neighbouring points, say: $x \text{ and } x + \Delta x$
- Next consider the difference between the squared distance between neighbours before and after deformation (with the appropriate normalization):

$$\frac{(f(x+\Delta x) - f(x))^2 - \Delta x^2}{2\Delta x^2}$$

- Finally take then limit as $\Delta x \to 0$ and use the Taylor approximation

$$E(x) = \frac{f'(x)^2 - 1}{2}$$

Deformation in 1D

- The mapping function should satisfy certain mathematical properties that make it physically "admissable" or "valid"
- It should be differentiable and should not map more than one point to the same final location. It should also preserve the order of points in the bar (easy to see why in 1D)
- In general, the more physical quantity, displacement is preferred: $q(x) = f(x) - x$
- In terms of this quantity the deformation can be written as: $E(x) = q_x + \frac{1}{2} q_x^2$
- And when the derivative of the displacement is small, can be linearized: $e(x) = q_x$

We chose $q(x)$ for the displacement because we want to reserve $u(x)$ for the velocity in fluid mechanics

Traction in 1D

- Once we have a way to quantitatively describe the deformation (often this is called STRAIN) we need a way to relate this to the forces in the interior of the continuum (the rod).
- What is the unique force in a continuum? Well it must be different from the so called body forces of particle mechanics (like gravity).
- It occurs in the interior of the continuum (0,*L*) and is due to neighbouring particles being moved from an equilibrium position. We will call it: $\tau(x)$
- This is much like the ideal, or Hookean, spring, and indeed the idea of elastic forces, which we will call TRACTION or STRESS, goes back to Robert Hooke in the 1600s.
- When the strain is small the simplest relation between stress and strain is a linear one:

$$\tau(x) = Ke(x)$$

Traction in 1D

- There is one nagging issue and that is sign convention.
- In 1D this is pretty easy, though in 3D it will come back and be non-trivial.
- We would like a positive force at the left hand endpoint of the rod to yield positive displacements.
- Say we clamp the right end, or u(L)=0, and integrate the traction using the definition of strain: $$\int_0^L \tau(x)\ dx = \int_0^L Kq_x\ dx = K[q(L) - q(0)]$$
- If K and u(0)>0 then we see that: $$\int_0^L \tau(x) dx < 0$$

Equilibrium in 1D

- Once we have the concepts of stress and strain we can complete our simplest "field theory" by deriving an equation for equilibrium when stress, or traction is the only force.
- We need the analogue of "sum of forces" from particle mechanics.
- Again consider neighours: $x \text{ and } x + \Delta x$
- The force per unit length is: $[\tau(x + \Delta x) - \tau(x)]/\Delta x$
- And again take the limit as Delta x goes to zero to get: $\frac{d\tau}{dx} = K\frac{d^2q}{dx^2} = 0$

Equilibrium in 1D

- Say we wanted to include gravity what would change?
- Define the solid density as ρ_s and take gravity to act in the negative x direction.
- The balance of force per unit length reads:

$$[\tau(x+\Delta x)-\tau(x)]/\Delta x \;\; -\rho_s g = 0$$

- Notice that since gravity is a body force there is no Delta x on the bottom, so in the limit: $K\dfrac{d^2q}{dx^2} - \rho_s g = 0$

Equilibrium in 1D: BVP

- Let's solve the simplest problem: The rod is clamped at x=L and a force, F, is specified at x=0. Recall that according to our sign convention:

$$\tau(0) = -F$$

- Integrating the no gravity equilibrium ODE gives: $q(x) = Ax + B$
- The BC at x=L means B=-AL
- The BC at x=0 means:

$$KA = -F$$

- And finally the solution reads: $q(x) = \frac{F}{K}(L - x) = \frac{F}{KL}\left(1 - \frac{x}{L}\right)$
- If F>0 then u>0 everywhere, as we expect on physical grounds.

Conclusions

- The 1D elastic problem is a lot narrower than the diffusion problem we derived first.
- But the two have interesting commonalities.
- The most important commonality is the way we had to relate the stress and the strain, and the way we had to relate the concentration and the flux.
- This is an example of a constitutive law, and it comes up over and over again in continuum mechanics.
- Constitutive laws have mathematical aspects, but fundamentally are a science problem. The difference between silly putty and steel comes down to constants that must be measured.
- Of course solids are simpler to describe than fluids, due to the fact that fluids flow.

More complex constitutive laws

- For an elastic material we are saying that the deformation is based on the spatial changes in displacement.
- This deformation is in turn taken to be proportional to the force due to neighbours, or traction.
- This basically means instantaneous adjustment, but many natural materials (like the silly putty I showed pictures) of actual take time to adjust.
- For such materials it might make sense to write something like $\tau(x,t) = \int_{-\infty}^{t} G(\tau)e(x,t-\tau)\ d\tau$. Here we have a different constant of proportionality for each time in the past.

  This type of integral is called a convolution
- If we imagine that memory fades away a very simple model might be $G(\tau) = G_0 \exp(-\tau/\tau_0)$. The parameter τ_0 tells you have far in the past we remember.

Viscoelastic Equilibrium in 1D

- The notion of equilibrium for a solid with memory, or a viscoelastic solid, doesn't change: $\tau_x - \rho_s g = 0$ but now the problem is that our relation to u is an integral.
- We can get around this by using the Laplace transform from ODE courses to write $\bar{\tau}(x,s) = \bar{G}(s)\bar{e}(x,s)$ where the overbars denote Laplace transformed quantities and we used the so-called convolution theorem to turn the funny integral into a multiplication in Laplace transformed space.
- The equilibrium condition in Laplace transformed space now looks like this

$$\bar{G}(s)\bar{q}_{xx} - \rho_s g = 0.$$

- So the idea is to solve in Laplace transformed space and then transform back.

What about fluids?

- The viscoelastic solid theory was still built on the idea that the basic way to measure deformation is based on displacement and how it varies in space.
- But the basic tenet of what a fluid is, that if we force a fluid it responds by flowing, so that it is no longer the displacement that we care about, but the velocity.
- So the basic variable is this velocity, $u = \dfrac{\partial q}{\partial t}$
- It is an unfortunate fact that many books denote both solid displacement and the fluid velocity by u. For this reason we have used q for displacement.
- The task of developing a 1D description of a moving fluid is taken up in the next lecture.